

Arhan Boyd

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EDUCATION

National University of Singapore

Singapore | January 2027 – Expected 2031

Doctor of Philosophy (PhD), Information Systems & Analytics, School of Computing (Incoming)

Awarded the NUS Research Scholarship (full tuition fee subsidy and monthly stipend)

Barcelona School of Economics

Barcelona, Spain | September 2023 – July 2024

Master's Degree in Specialized Economic Analysis, Program in Competition, Regulation and Markets

Degree awarded by Universitat Pompeu Fabra and Universitat Autònoma de Barcelona

Specialization – Competition Economics, Mergers, Regulatory Economics, Economics of AI, Game Theory

Thesis – The Impact of Data Privacy Regulations through a Cournot Modelling Approach

University of Queensland

Brisbane, Australia | July 2020 – July 2023

Bachelor of Economics (B.Econ), Major in Quantitative Analysis / Mathematics

Specialized Area of Interest – Game Theory, Advanced Microeconomics, Machine Learning, Deep Learning, Computational Economics, Data Science, Operations Research

PUBLICATIONS

- Smerdon, D., Boyd, A., Reizniece, D., and Sorokina, A. (2026). 2026 FIDE Gender Equality in Chess Index (GECI). Report, FIDE and The University of Queensland. DOI: 10.14264/9ec1c7e.

WORKING PAPERS

- "David vs Goliath: Who Wins under GDPR" (with Shiva Shekhar and Debapratim Ghosh). Submitted to the Workshop on Information Technology and Systems (WITS).
- "What Can Digital Payment Data Tell Us About India's Economy? Evidence from PhonePe Pulse" (with Ravi Bapna, Vandith Pamuru, and Jonathan Hersh). Accepted at the Conference on Information Systems and Technology (CIST) 2026.
- "Weak Monotonicity for Implementability in Max Domains" (with Souvik Roy and Satish Aggarwal). Working paper.
- "The Impact of Data Privacy Regulations through a Cournot Modelling Approach" (with Souvik Roy). (Master's thesis; in progress to be a Working Paper).

WORK IN PROGRESS

- Follow-up project on digital payments and India's digital economy using more granular PhonePe Pulse data (with Ravi Bapna, Vandith Pamuru, and Jonathan Hersh). Title to be determined.
- Healing Networks: an independent NLP and network theory project on emotional recovery after heartbreak, using sentiment analysis and BERT models on Reddit, X and LinkedIn (mentored by Ravi Bapna).

CONFERENCE PRESENTATIONS

- Conference on Information Systems and Technology (CIST) 2026 – "What Can Digital Payment Data Tell Us About India's Economy? Evidence from PhonePe Pulse" (forthcoming).
- NCAER Workshop on the Economics of AI and Digital Public Infrastructure, New Delhi (June 2026) – PhonePe Pulse research presented.
- Lear Competition Festival, Rome (2024) – "The Impact of Data Privacy Regulations through a Cournot Modelling Approach" (Master's thesis).

MEDIA COVERAGE

- Research on PhonePe Pulse cited in Sabharwal, M. and Chandra, K., "Enterprise DPI can help India reclaim its advantage in the AI game," Business Standard, 23 July 2026.

PROFESSIONAL EXPERIENCE

Indian School of Business

Hyderabad, India | August 2025 – Present

Research Associate (Information Systems)

Supervisors: Vandith Pamuru (ISB), Ravi Bapna (University of Minnesota)

- Research Assistant on a paper by Ravi Bapna and Gordon Burtch studying the effects of Name, Image and Likeness (NIL) policies on college athletes' athletic performance; responsible for addressing referee comments, scraping additional data, refining methodologies, additional econometric analysis, and improving the paper for resubmission.
- Working with Vandith Pamuru on auctions research: undertook a literature review on combinatorial auctions, presented reviews of several seminal papers on Generalized Second-Price (GSP) auctions, and currently assisting on a project studying auction design and bidding in blockchain-based NFT marketplaces.
- General research assistance across projects, including literature reviews, data collection and cleaning, web scraping, econometric and causal analysis, programming, and preparing research reports and presentations.

Barcelona School of Economics

Barcelona, Spain (Remote) | September 2024 – July 2025

Research Analyst (Contract)

- Worked under the supervision of Amedeo Piolatto and Martin Besfamille on a project examining municipal union formation in Italy, focusing on how heterogeneity in state capacity affects incentives to collaborate on local public good provision. The work involved data wrangling, economic modelling, and econometric analysis to identify causal pathways for union formation.

- Collaborated with Benito Arruñada on a research project applying transformer-based NLP models (BERT, Hugging Face) and sentiment analysis to historical texts of Dominican and Franciscan traditions. The project investigated how theological doctrines shaped early economic behaviors, emphasizing differences in concepts like charity, justice, rationality, and ownership across the two orders. Responsible for model fine-tuning, preprocessing large historical corpora (Latin and English), and building interpretability layers to analyze ideological differences and their link to institutional design.

University of Queensland Brisbane, Australia | January 2021 – August 2023
Research Assistant (Research Analyst), School of Economics

- Supervisors** – Valentin Zelenyuk, Allan H Chanto, Flavio Menezes, Metin Uyanik, David Smerdon
- Led a Nash Bargaining project, applying advanced mathematical modelling to real-world scenarios.
 - Managed large-scale administrative and experimental datasets and applied machine learning and econometric models to improve predictive reliability.
 - Lead Research Assistant to Allan Hernandez, managing a team of 7 on data collection and analysis for second-price oil auctions in the United States.
 - As a GCYR Scholar (secondment to the Australian Institute of Business & Economics, 2022), delivered a research report on Australia's energy transition under Flavio Menezes.
 - As a Winter Research Scholar under David Smerdon, independently designed and ran primary survey data collection from international sources, including outreach to officials across several countries, with web scraping, cleaning, and analysis.
 - Contributed to ARC Grant-funded productivity and efficiency research under Valentin Zelenyuk using Data Envelopment Analysis; work acknowledged in two peer-reviewed publications.

University of Queensland Brisbane, Australia | February 2022 – June 2023
Teaching Assistant, School of Economics

- As an academic tutor my responsibilities include teaching a class of 20–30 students, marking papers, and preparing tutorial worksheets.
- **Subjects Taught** – Advanced Microeconomics, Game Theory, Intermediate Microeconomics

Indian Statistical Institute Kolkata, India (Remote) | March 2021 – June 2023
Research Assistant

- Assisted Dr. Souvik Roy (ISI, Kolkata) on multiple projects in game theory and mechanism design: literature reviews, research reports, coding, mathematical modelling, and simulations.
- **Research Interests** – Applied Game Theory, Microeconomic Theory, Mechanism Design, Cheap Talk

ACHIEVEMENTS / AWARDS

- NUS Research Scholarship (2026), awarded with admission to the PhD programme at the School of Computing, National University of Singapore.
- Building a quantitative finance model for macro regime classification (Goldilocks, Reflation, Stagflation, Deflation), combining macroeconomic and market data with LLM-based news sentiment to generate forward-looking regime probabilities.
- Honorary Advisor (AI & Competition) @ GW Competition & Innovation Lab, George Washington University (October 2024 – March 2025).
- GCYR Research Scholar (2022), Summer Research Scholar (2022), Winter Research Scholar (2021).
- Deans Commendation for Academic Excellence (2021).
- Australian Universities Chess League Champions (2021) & National Teams Chess Champion (2019).

INDUSTRY EXPERIENCE

CampusBinge (ISB-incubated startup) January 2026 – Present
AI Consultant

- Helped design and build Lattice, the startup's AI model for college and career guidance, covering LLM orchestration, retrieval over verified data, and reasoning and simulation layers.

Boyd Smiths Pvt Ltd. December 2024 – August 2025
Advisor (Automation)

- Spearheaded automation and AI-based modelling for production planning, workforce and machine optimization at a manufacturing unit.

ADDITIONAL SKILLS

Technical Skills
Python, R, Matlab, Julia, Stata, Latex, Tensorflow, MS Excel, SPSS, Power BI, SQL, Qualtrics

Industry Skills
Data Analysis, LLMs, Transformers, Prompt Engineering, Natural Language Processing, Econometrics, Causal Inference, Machine Learning, Deep Learning, Research, Programming, MS Excel, Forecasting, Predictive Modelling, Data Visualization, Data Science, Cost-Benefit Analysis, Text Analysis, Data Wrangling, Mergers & Acquisitions, Competition Economics, Economic Consulting

Languages
English (Native), Hindi (Native), Bengali (Advanced), Spanish (Basics)

EXTRA-CURRICULAR & INTERESTS

- Advanced Chess Player (Tournament Level)
- Poker Player (Intermediate)
- Water Polo (Club Level)